

A culture-independent sequence-based metagenomics approach to the investigation of an outbreak of Shiga-toxigenic *Escherichia coli* O104:H4.

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Evidence abstract

IMPORTANCE Identification of the bacterium responsible for an outbreak can aid in disease management.

However, traditional culture-based diagnosis can be difficult, particularly if no specific diagnostic test is available for an outbreak strain.

OBJECTIVE To explore the potential of metagenomics, which is the direct sequencing of DNA extracted from microbiologically complex samples, as an open-ended clinical discovery platform capable of identifying and characterizing bacterial strains from an outbreak without laboratory culture.

DESIGN, SETTING, AND PATIENTS In a retrospective investigation, 45 samples were selected from fecal specimens obtained from patients with diarrhea during the 2011 outbreak of Shiga-toxigenic *Escherichia coli* (STEC) O104:H4 in Germany.

Samples were subjected to high-throughput sequencing (August-September 2012), followed by a 3-phase analysis (November 2012-February 2013).

In phase 1, a de novo assembly approach was developed to obtain a draft genome of the outbreak strain.

In phase 2, the depth of coverage of the outbreak strain genome was determined in each sample.

In phase 3, sequences from each sample were compared with sequences from known bacteria to identify pathogens other than the outbreak strain.

MAIN OUTCOMES AND MEASURES The recovery of genome sequence data for the purposes of identification and characterization of the outbreak strain and other pathogens from fecal samples.

RESULTS During phase 1, a draft genome of the STEC outbreak strain was obtained.

During phase 2, the outbreak strain genome was recovered from 10 samples at greater than 10-fold coverage and from 26 samples at greater than 1-fold coverage.

Sequences from the Shiga-toxin genes were detected in 27 of 40 STEC-positive samples (67%).

In phase 3, sequences from *Clostridium difficile*, *Campylobacter jejuni*, *Campylobacter concisus*, and *Salmonella enterica* were recovered.

CONCLUSIONS AND RELEVANCE These results suggest the potential of metagenomics as a culture-independent approach for the identification of bacterial pathogens during an outbreak of diarrheal disease.

Challenges include improving diagnostic sensitivity, speeding up and simplifying workflows, and reducing costs.

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